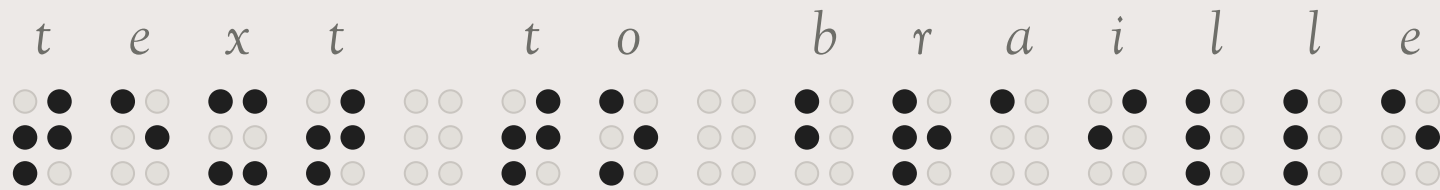


TEXT-TO-BRAILLE

Transform sight into touch.

t e x t t o b r a i l l e



Ebrahim Udaipurwala

Luca Aquilani

Richard Simorangkir

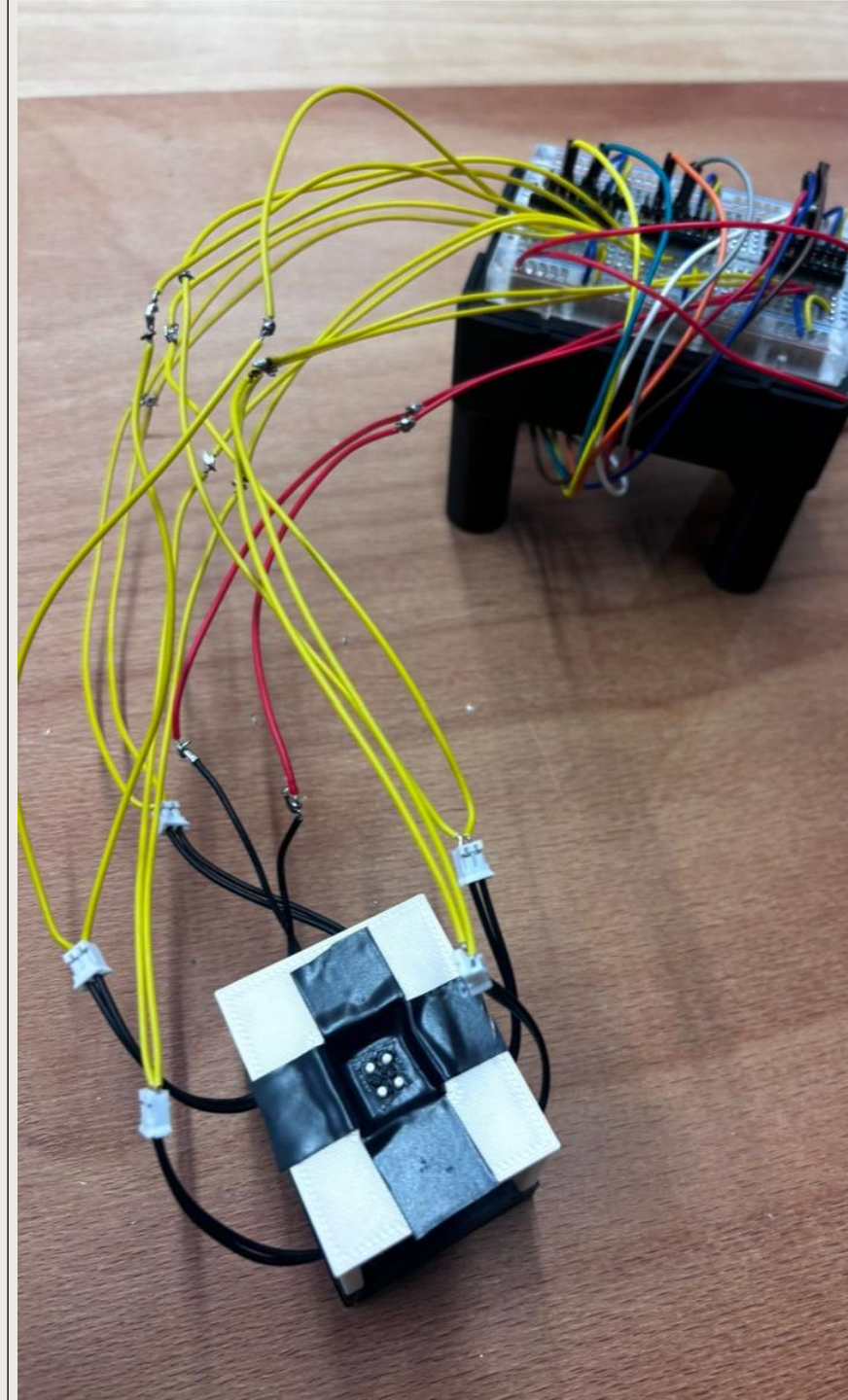
Arihan Gupta

Ansar Ahmad

Marcel Zuhmann

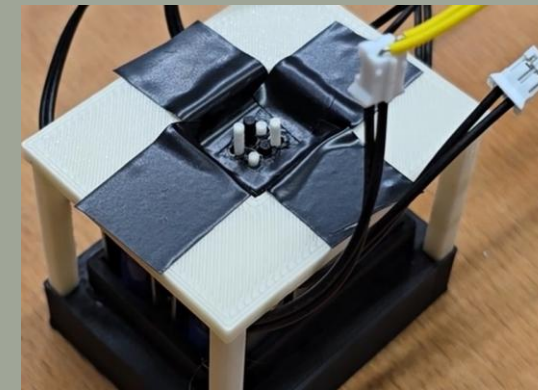
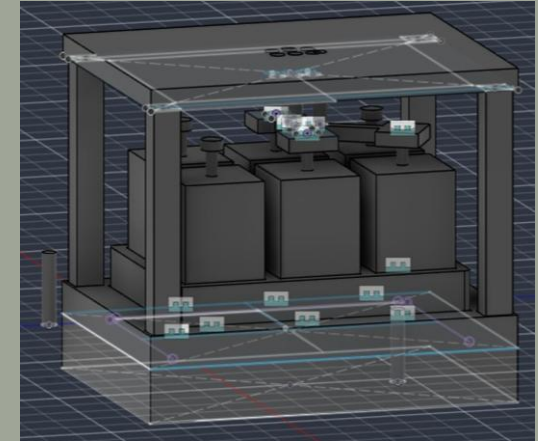
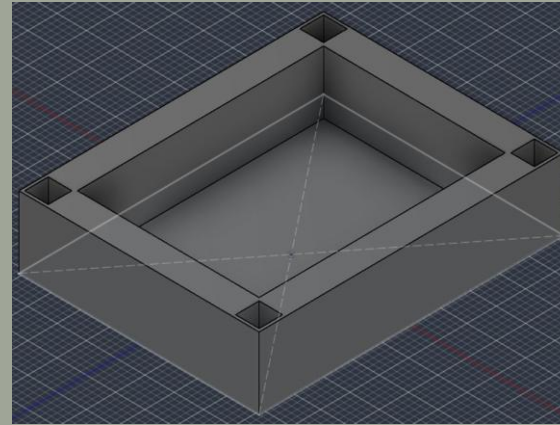
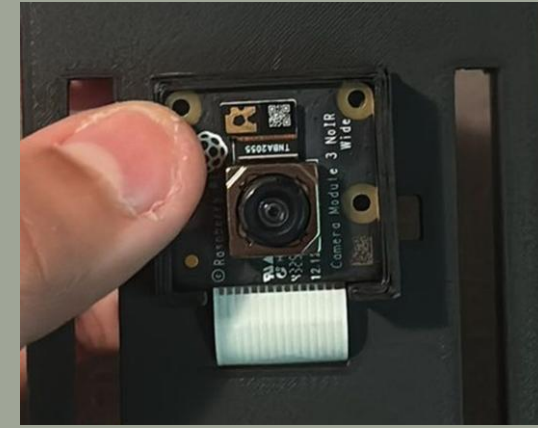
Alexandru Oprea

Timofei Rusanov



CONTENTS

- What is Braille?
- Project Introduction
- Image $\rightarrow$ Text
- Text $\rightarrow$ Braille
- Physical Circuit & Model
- Braille $\rightarrow$ Physical Pins
- Evaluation
- Future Work



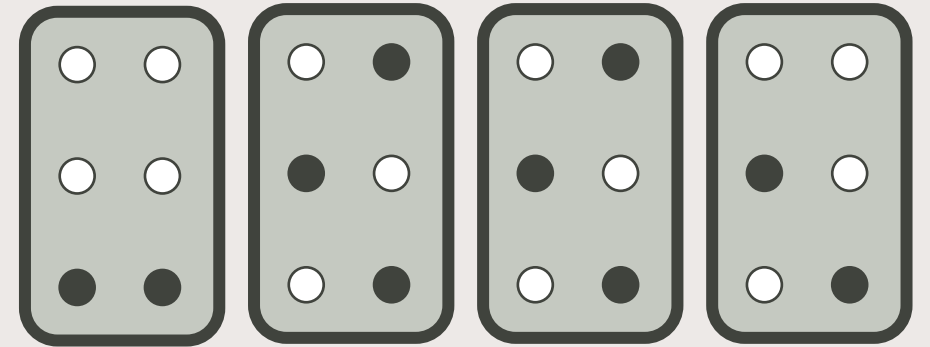
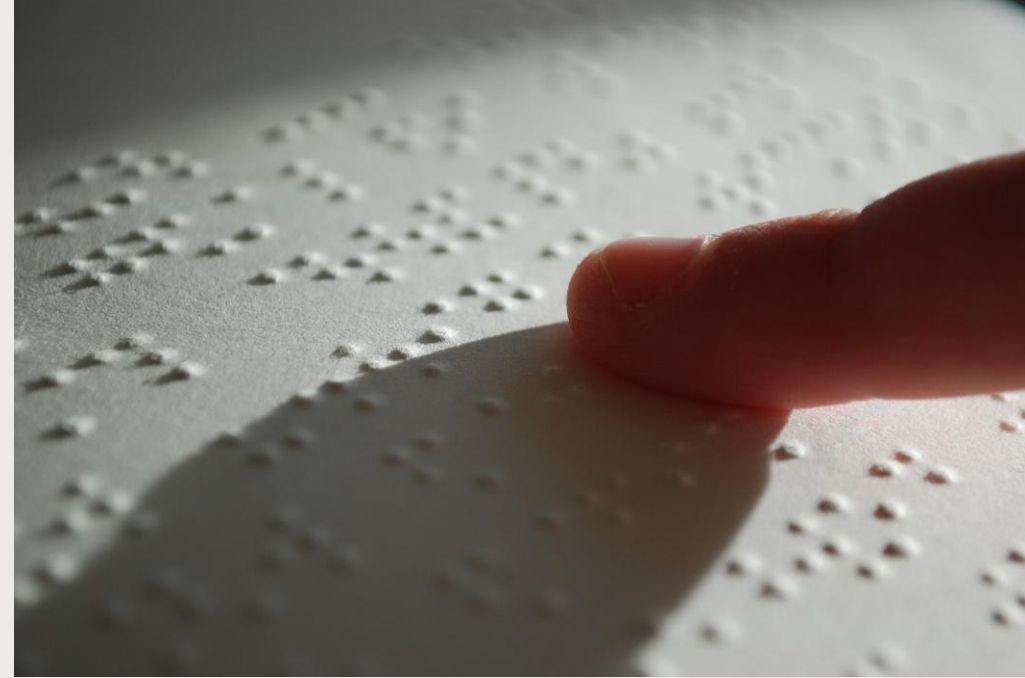
What is Braille?

Tactile writing
system for blind
and visually
impaired people

Uses raised dots
arranged in cells

Each cell has 6
dots in different
patterns

Represents letters,
numbers, and
symbols



Text to Braille



T e x t t o B r a i l l e

PRINTED TEXT IS STILL OUT OF REACH



Printers are rare

Special embossers, found in very few places.



Nothing portable

Displays stay on a desk and need digital text first.
Not readily available



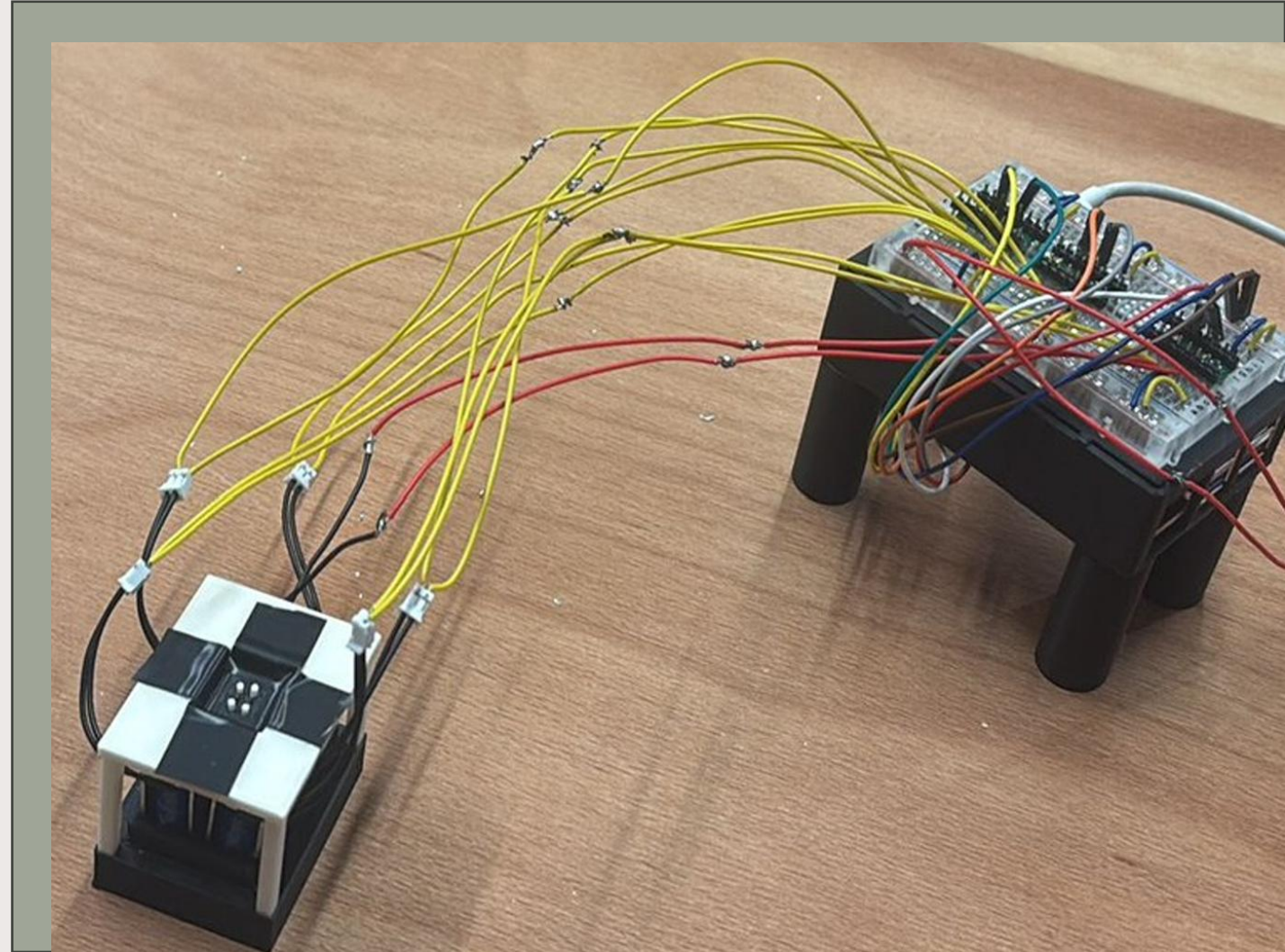
Speech isn't reading

Text-to-speech is slow and hides spelling and layout.

OUR PROJECT – TEXT TO BRAILLE

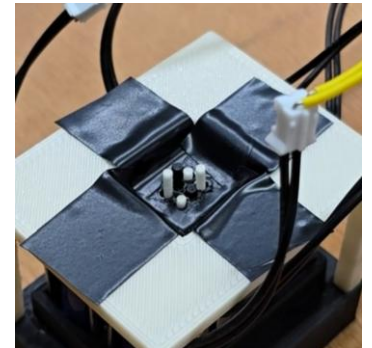
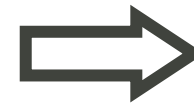
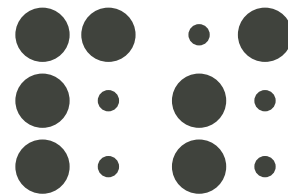
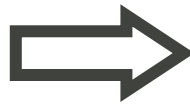
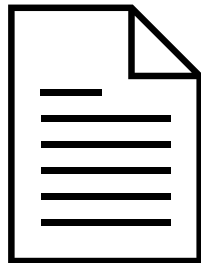
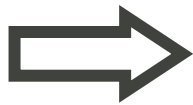
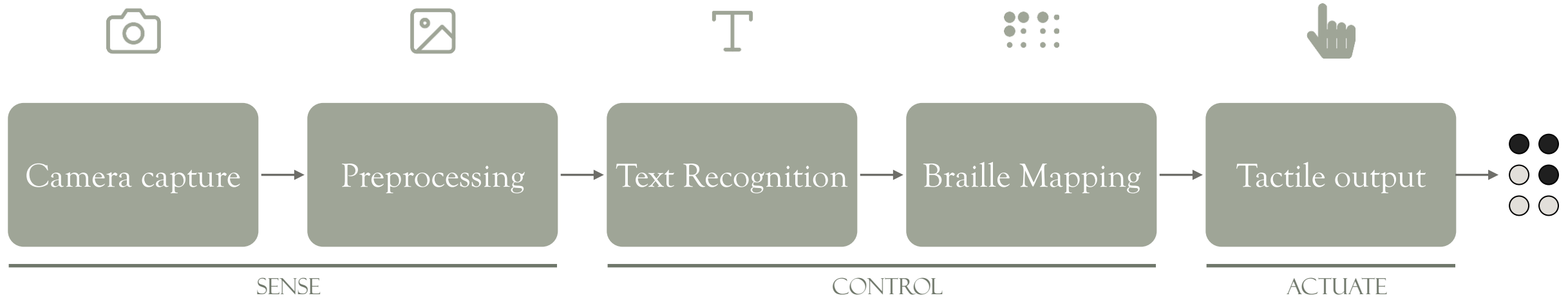
- Tactile replacement of text-to-speech
- Support Braille education

- Refreshable Braille Device
 - Equipped with a camera
1. Take a picture of the document
 2. Translate into Braille
 3. Displayed dynamically at a configurable speed



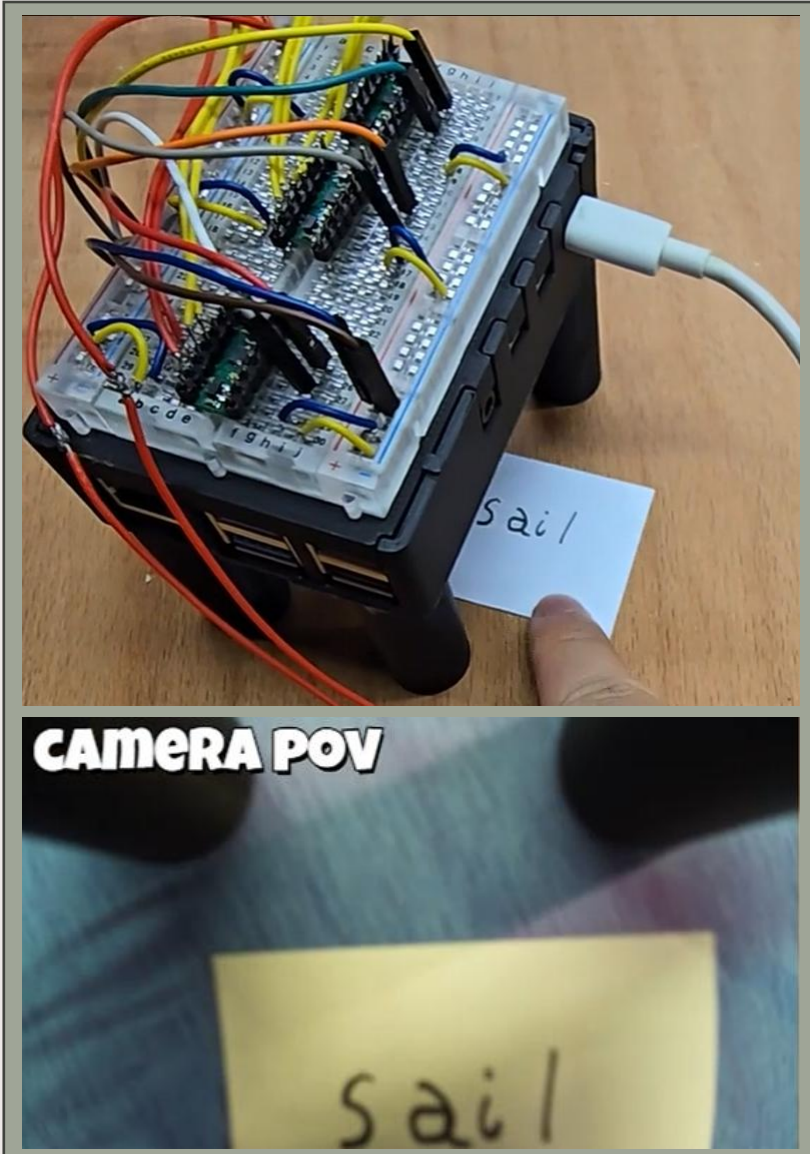
ONE LOOP

SENSE, CONTROL, ACTUATE



VIDEO DEMO

FROM IMAGE TO TEXT - OCR



Input: Live Camera Feed and Still Picture



Picture is processed by an OCR API



The Raspberry Pi 4 can also run the OCR locally.



Live view for framing

Small preview stream with status and the last recognized text.



Output: Digital text

OCR - STATISTICS

99.4%

Cloud OCR accuracy

OCR.space, normal light

96.4%

In dim lighting conditions

≈1.6s

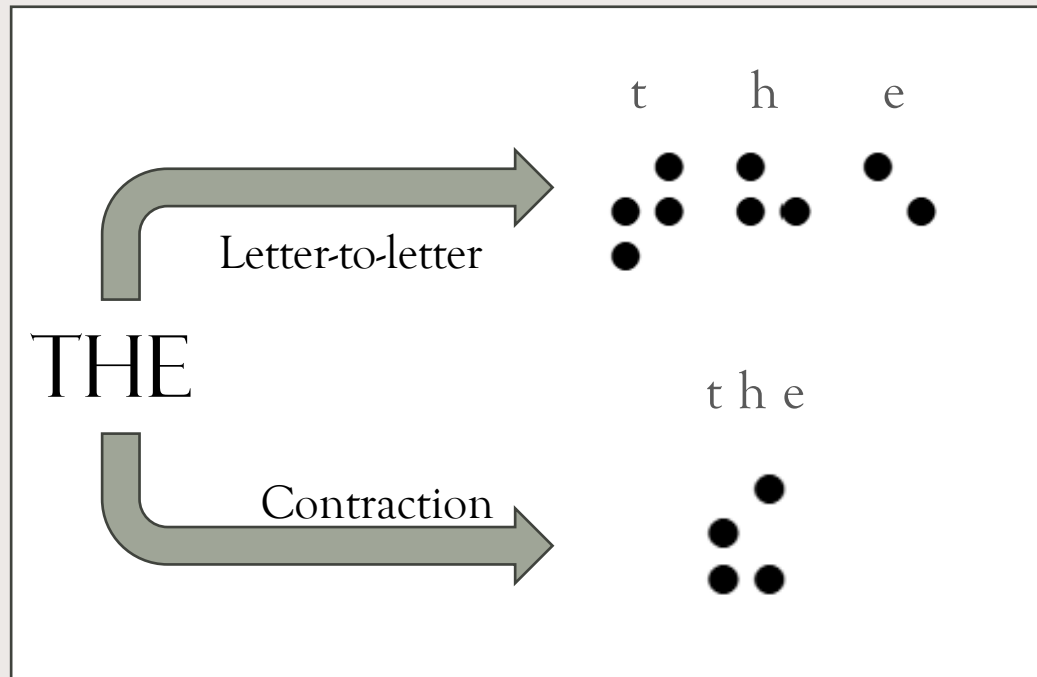
Scan to output latency

key press to tactile output

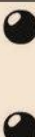
- Works accurately in different light conditions

FROM TEXT TO BRAILLE

- Direct conversion for each letter
 - *shortcuts are used as well!*



Braille Alphabet

a	b	c	d	e
				
f	g	h	i	j
				
k	l	m	n	o
				
p	q	r	s	t
				
u	v	w	x	y
				
z	period	comma	Braille is the system of six raised dots created in 1821 by French schoolboy Louis Braille. It is the only medium through which children with profound or total loss of sight can learn to read and write.	
				

CONVERSION

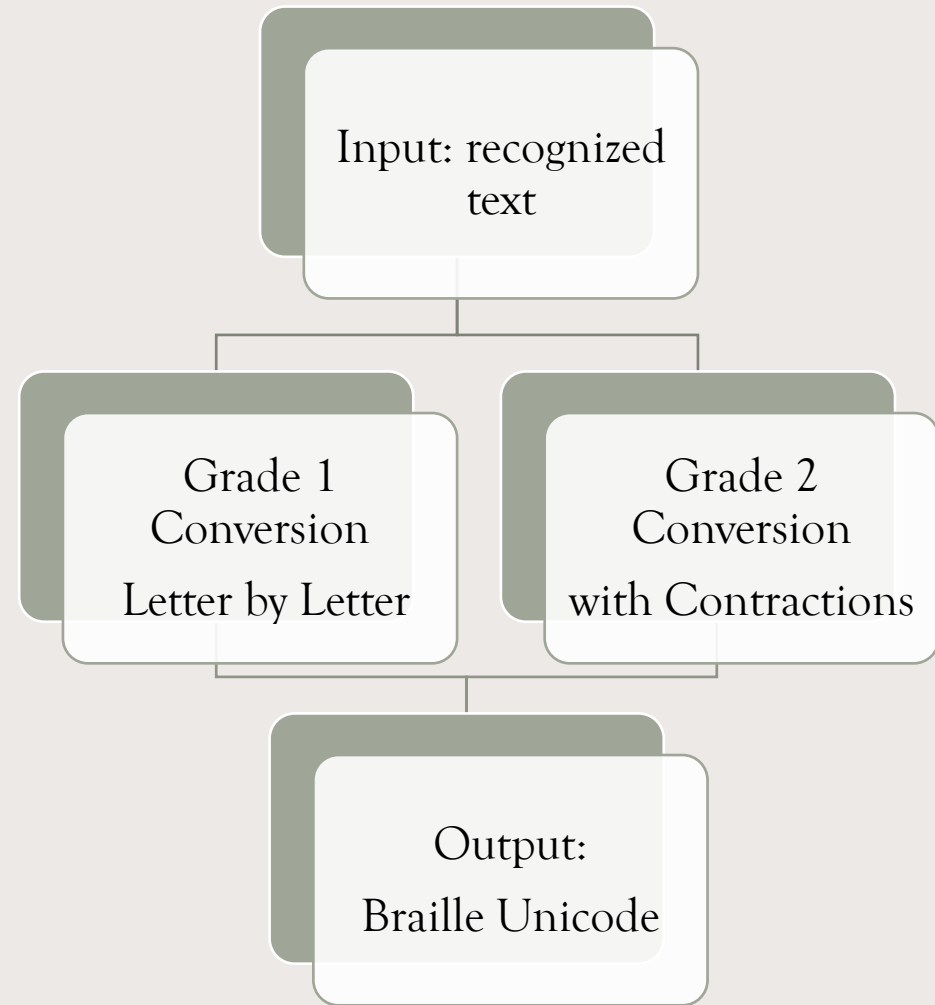
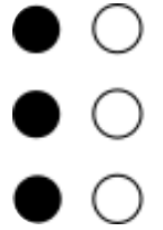
TEXT → BRAILLE

Liblouis library

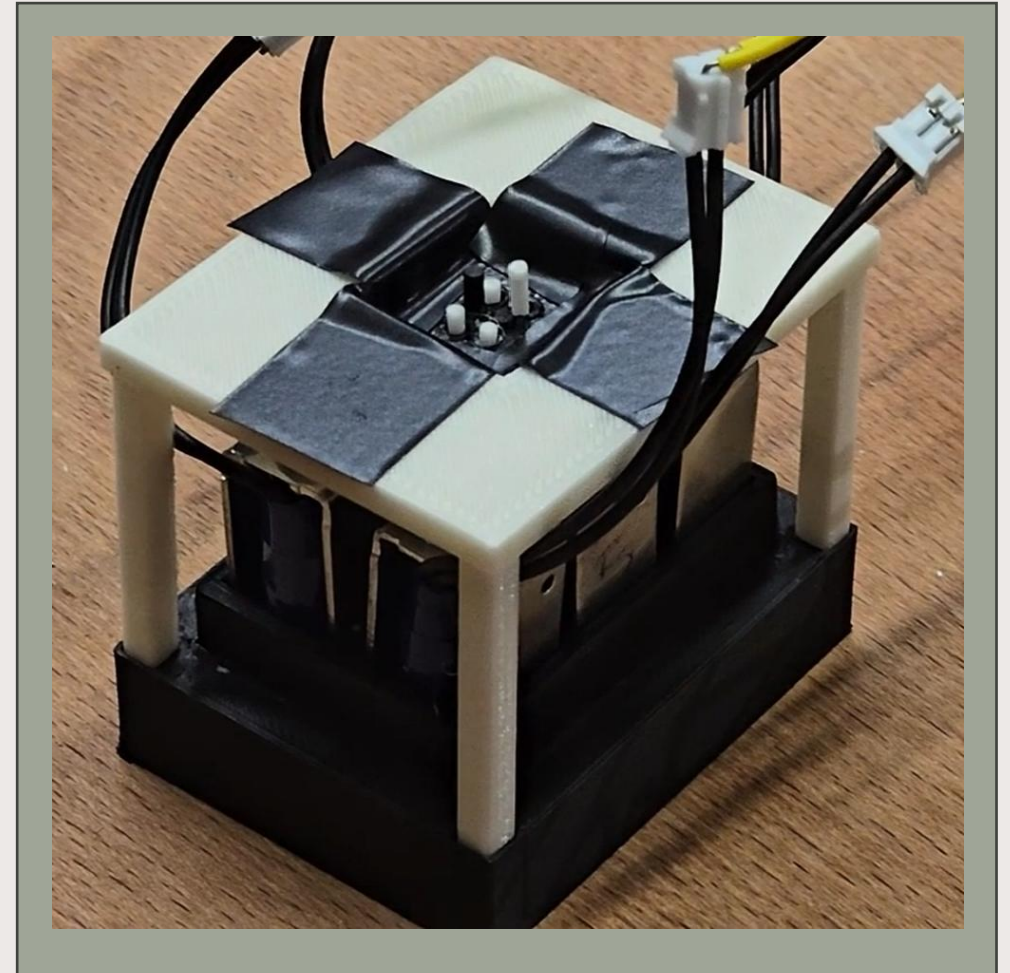
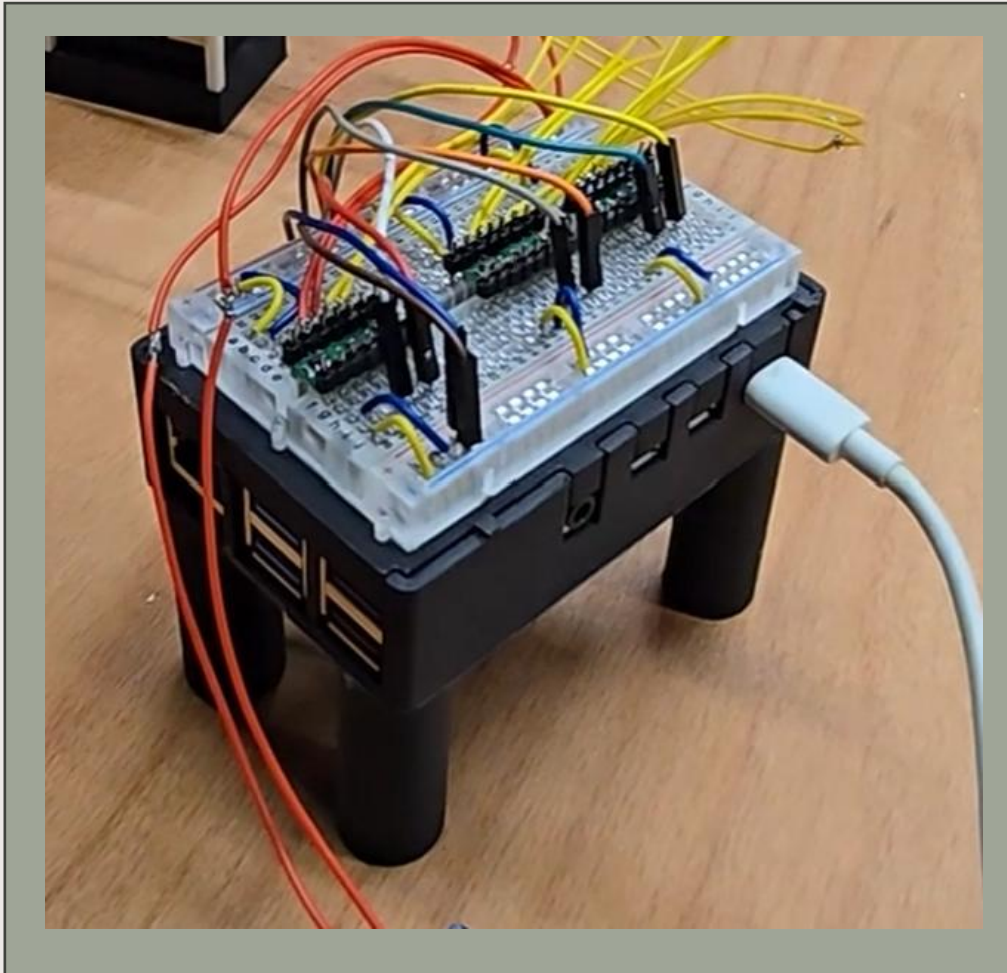
- Python library for Braille work
- Automatically detects contractions

liblouis/**liblouis**

Open-source braille translator and back-translator.



THE MODEL



THE MODEL



Scanning unit

Camera and control devices.



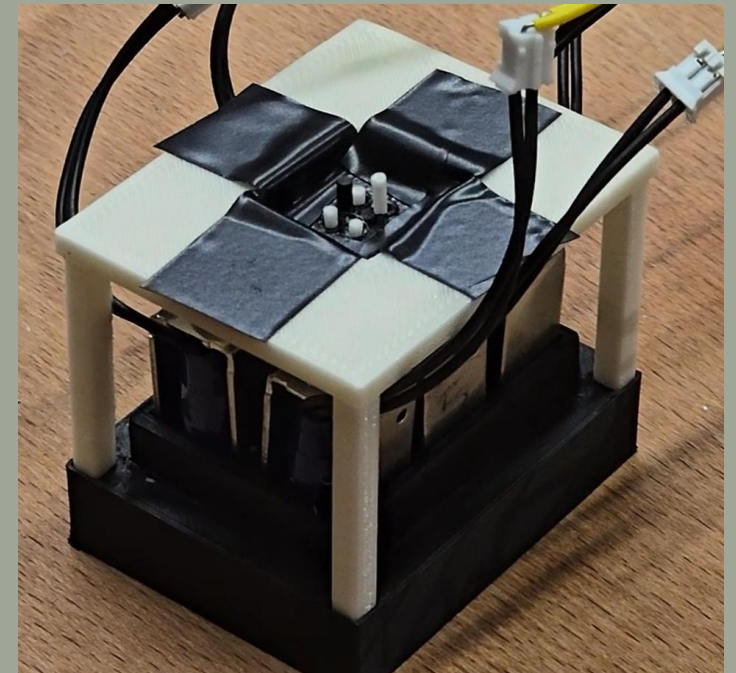
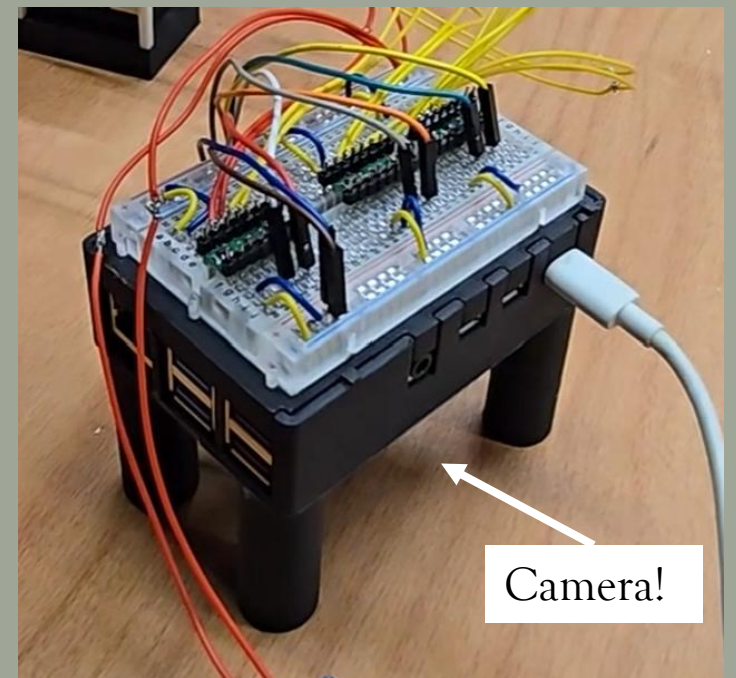
Reading unit

Six pins, 2.5 mm spacing: standard Braille under one fingertip.

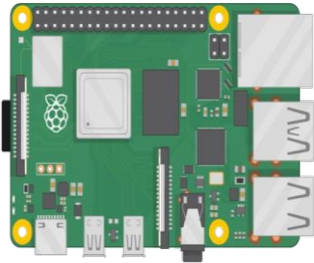


Built to be handled

Camera is firm and stable. Case opens for maintenance.



TECHNICAL COMPONENTS



Raspberry Pi 4 B

Runs capture, translation and GPIO control



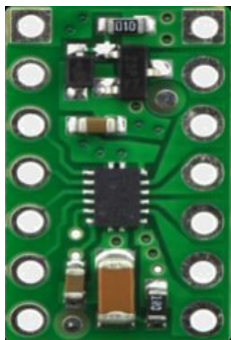
Camera Module

Captures the page at 2028×1520



6 × 5 V mini solenoids

One push-pull pin per Braille dot



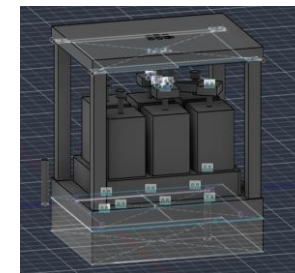
3 × Pololu DRV8835

Two driver channels each, six in total



USB-C breakout

Power for computation and actuation

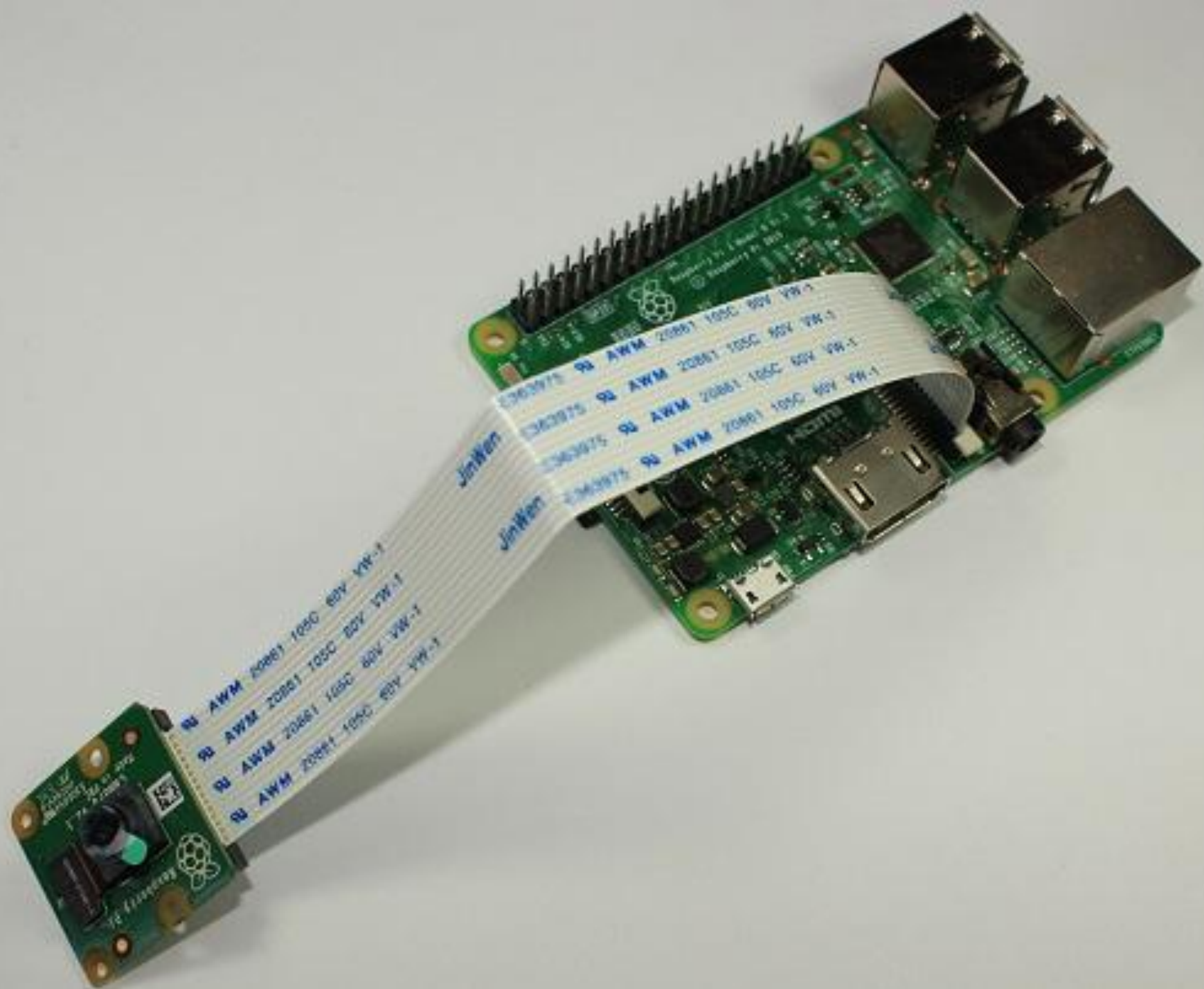


Printed case

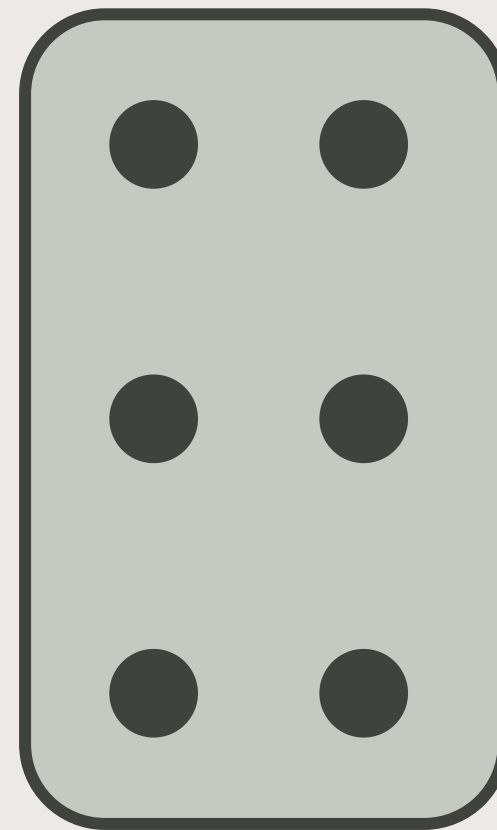
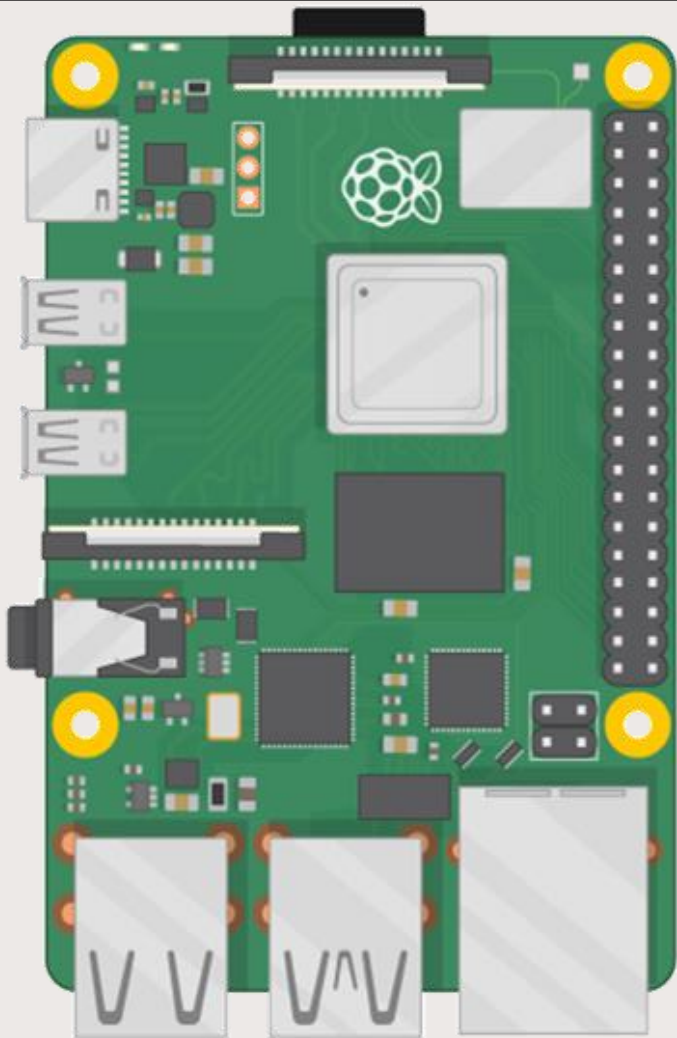
Two housings, designed in Fusion 360

CAMERA

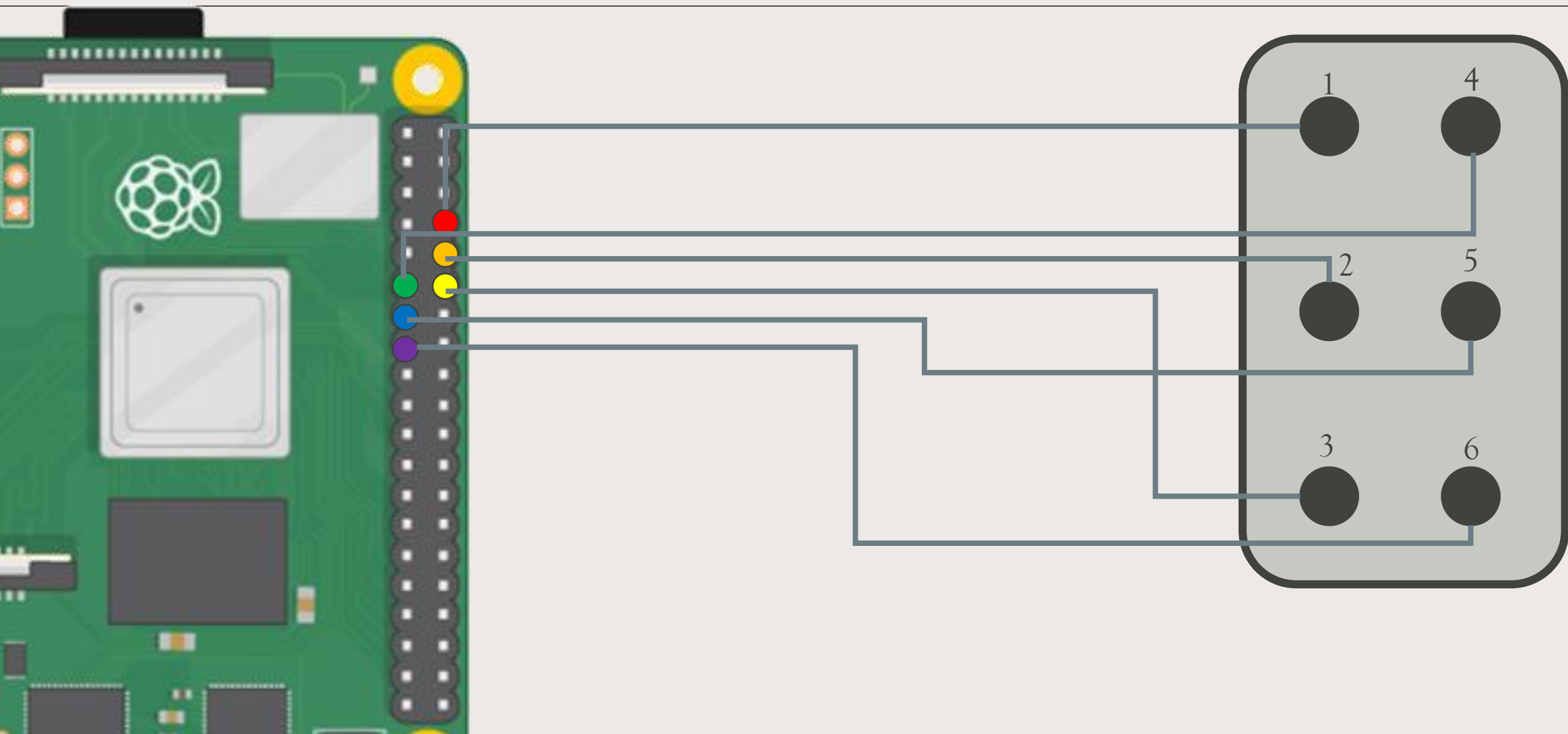
- Direct connection to Raspberry Pi
- No transmission delay



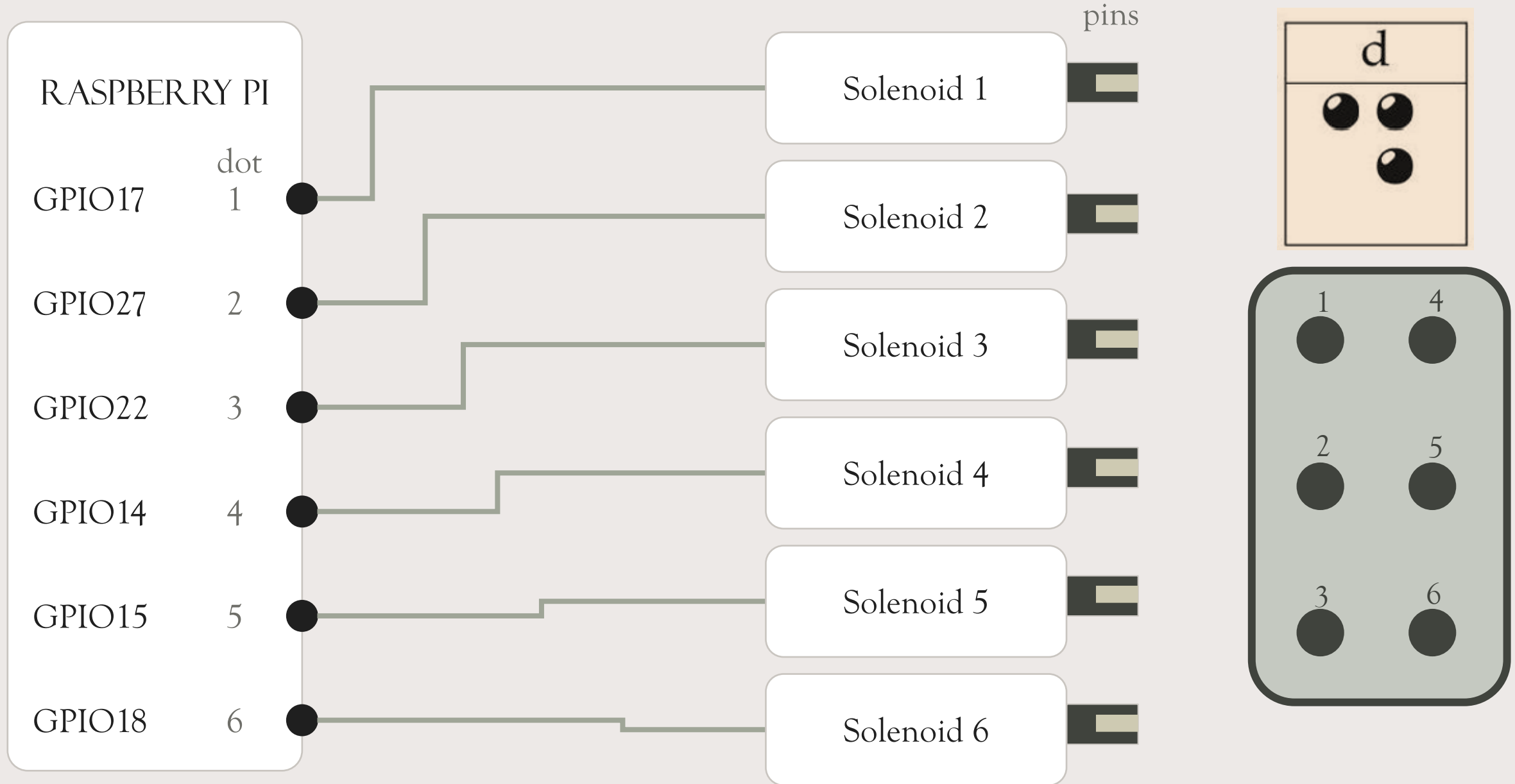
FROM BRAILLE TO THE PHYSICAL PINS



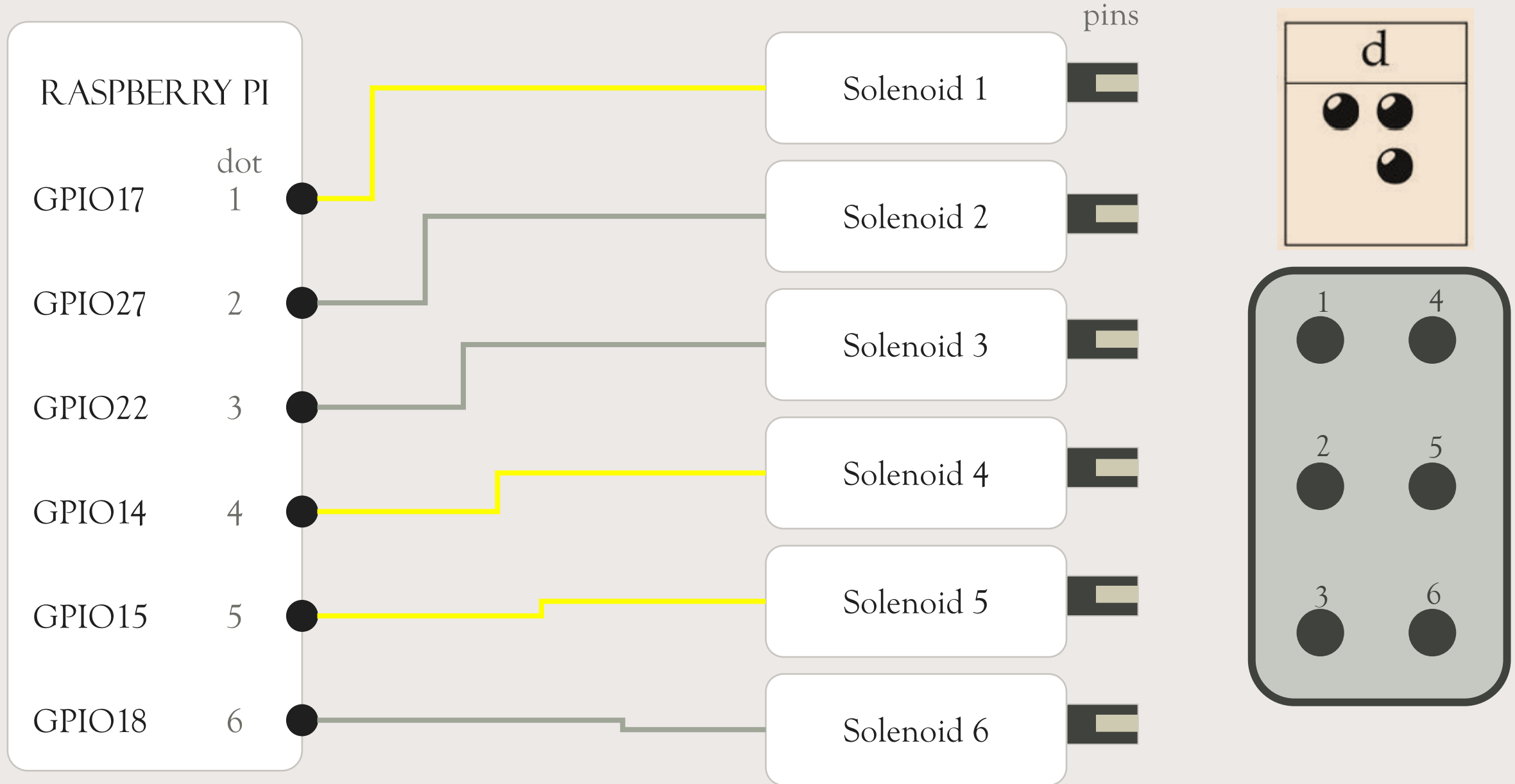
FROM BRAILLE TO THE PHYSICAL PINS



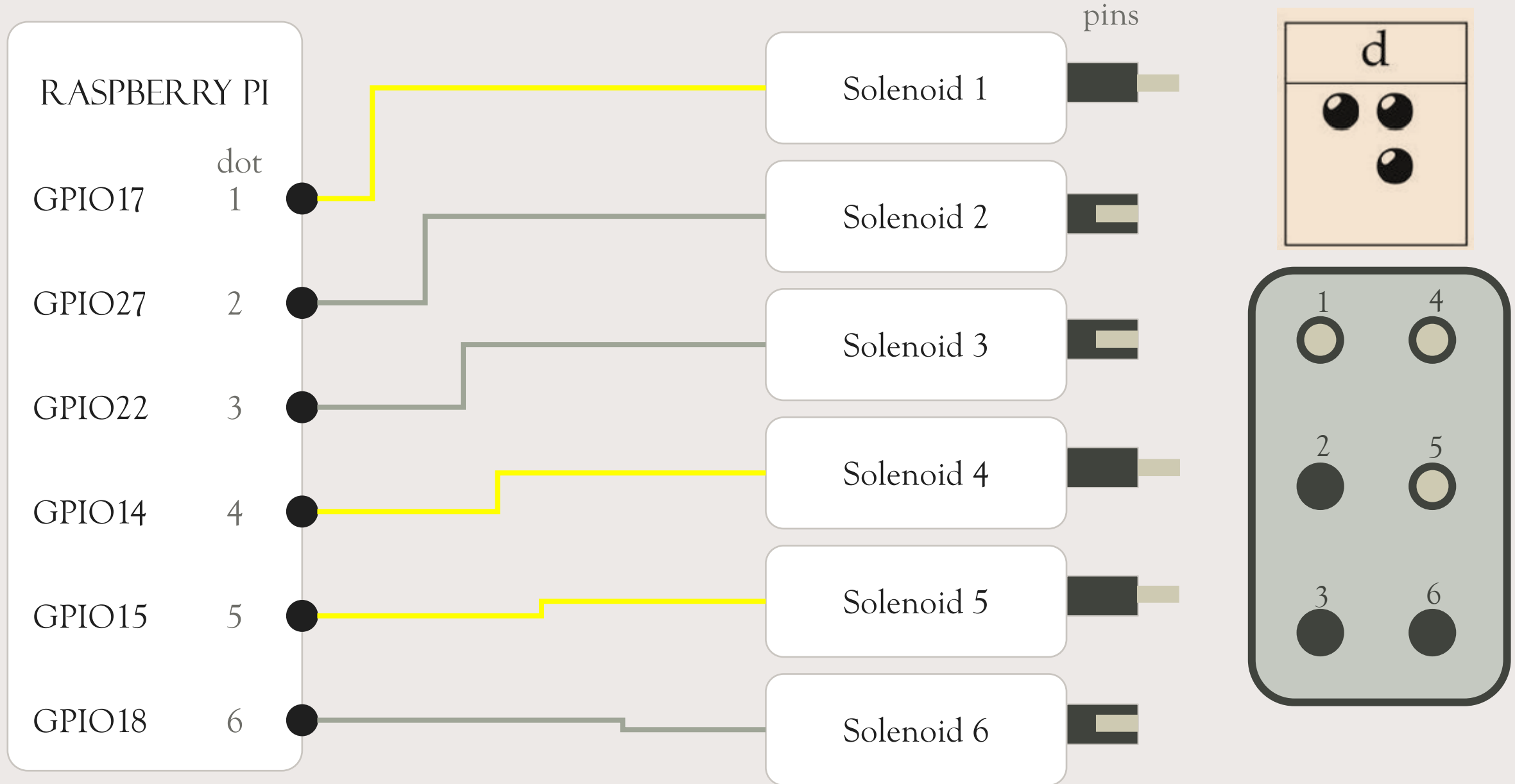
FROM BRAILLE TO THE PHYSICAL PINS



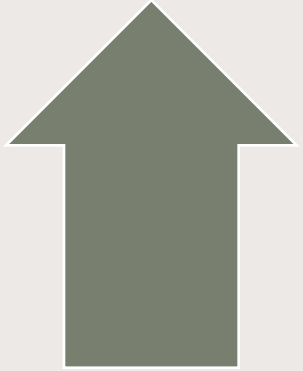
FROM BRAILLE TO THE PHYSICAL PINS



FROM BRAILLE TO THE PHYSICAL PINS



PROJECT EVALUATION



High Accuracy
& Low Latency

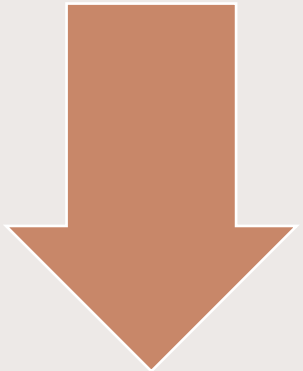
>95% in all cases | <1.8s

Support Braille

Education & Versatility

Physical separation

Scanning & Reading units



Components
overheat

Raspberry Pi
lacks in power

No user
validation

WHAT'S NEXT



More portability

Access to Braille text at any time on the go.
Battery powered.



Integrated finger module

Drivers next to the pins, no tether to the Braille (actuation) unit.



Real-time reading

Scan along the text, get instant feedback



Real users

Validation with Braille readers and learners.

THANKS



Questions?

THANKS

